

Investment and finance exam

Exercise 1

$$\bar{r} = \begin{pmatrix} 0.1 \\ 0.08 \\ 0.15 \end{pmatrix}$$
$$\Sigma = \begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix}$$

a)

$$w = \begin{pmatrix} 0.36 \\ 0.74 \\ -0.1 \end{pmatrix}$$

We calculate using standard formulas

$$\bar{r}_w = \bar{r}^T \cdot w = \text{transpose} \left(\begin{pmatrix} 0.1 \\ 0.08 \\ 0.15 \end{pmatrix} \right) \begin{pmatrix} 0.36 \\ 0.74 \\ -0.1 \end{pmatrix} \approx 0,0802$$

$$\sigma_w^2 = w^T \cdot \Sigma \cdot w = \text{transpose} \left(\begin{pmatrix} 0.36 \\ 0.74 \\ -0.1 \end{pmatrix} \right) \begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix} \begin{pmatrix} 0.36 \\ 0.74 \\ -0.1 \end{pmatrix} = 0,070684$$

b)

we are in the markowitz framework so we simply use the formula that we know gives exactly the minimum variance given wanted return in this world

$$w^* = \frac{a \cdot \bar{r}_p - b}{d} \Sigma^{-1} \bar{r} + \frac{c - \bar{r}_p b}{d} \Sigma^{-1} \mathbb{I}$$

With

$$a = \mathbb{I}^T \Sigma^{-1} \mathbb{I}$$
$$b = \mathbb{I}^T \Sigma^{-1} \bar{r}$$
$$c = \bar{r}^T \Sigma \bar{r}$$
$$d = ac - b^2$$

We use our r document from the coding bootcamp

Which gives us

> w_mark

```
      [,1]  
[1,] 0.3895131  
[2,] 1.0074906  
[3,] -0.3970037
```

I'll call this portfolio B

$$w_B = \begin{pmatrix} 0.3895 \\ 1.0075 \\ -0.3970 \end{pmatrix}$$

This asset has a variance

$$\text{transpose} \left(\begin{pmatrix} 0.3895 \\ 1.0075 \\ -0.3970 \end{pmatrix} \right) \begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix} \begin{pmatrix} 0.3895 \\ 1.0075 \\ -0.3970 \end{pmatrix} \approx 0,079176$$

And is thus a “better” asset than the portfolio described in (a)

Since it has the same return(roughly) and lower variance

c) again we have a formula for the weights of this portfolio

$$w_{tan} = \frac{\Sigma^{-1} \bar{r}_e}{\mathbb{I}^T \Sigma^{-1} \bar{r}_e}$$

$$\bar{r}_e = \bar{r} - r_f \cdot \mathbb{I} = \begin{pmatrix} 0.1 \\ 0.08 \\ 0.15 \end{pmatrix} - 0.02 = \begin{pmatrix} 0,08 \\ 0,06 \\ 0,13 \end{pmatrix}$$

$$w_{tan} = \frac{\begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix}^{-1} \begin{pmatrix} 0,08 \\ 0,06 \\ 0,13 \end{pmatrix}}{\text{transpose} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) \begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix}^{-1} \begin{pmatrix} 0,08 \\ 0,06 \\ 0,13 \end{pmatrix}} \approx \begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \end{pmatrix}$$

And we use standard formulas to find mean and variance

$$\bar{r}_{tan} = \text{transpose} \left(\begin{pmatrix} 0.1 \\ 0.08 \\ 0.15 \end{pmatrix} \right) \begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \end{pmatrix} \approx 0,1177056$$

$$\sigma_{tan}^2 = \text{transpose} \left(\begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \end{pmatrix} \right) \begin{pmatrix} 0.18 & 0.02 & 0.08 \\ 0.02 & 0.1 & 0.1 \\ 0.08 & 0.1 & 0.25 \end{pmatrix} \begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \end{pmatrix} \approx 0,1205459$$

d) I would recommend an efficient asset

from the one fund theorem we know that any efficient asset when there is a risk free asset must be a combination of the tangency portfolio and the risk free asset

thus we want to find the weights in the tangency portfolio and the risk free portfolio such that

$$0.06 = \alpha \cdot \bar{r}_{tan} + (1 - \alpha)r_f$$

$$0.06 = \alpha \cdot 0,1177056 + (1 - \alpha)0.02$$



Ligningen løses for α vha. WordMat.

$$\alpha = 0,4093931$$

$$1 - 0,4093931 = 0,5906069$$

Thus we must have 40.94% in the tangency portfolio and 59.06% in the risk free asset

This gives us a total portfolio with the 4. Asset being the risk free one

$$w_{efficient} = 0,4093931 \begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \\ 0 \end{pmatrix} + 0,5906069 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 0,1329198 \\ 0,09393 \\ 0,1825432 \\ 0,5906069 \end{pmatrix}$$

e) to show that it is on the SML we must first find its beta when CAPM is satisfied then

$$w_{tan} = w_m$$

variance and return also

we can calculate all 3 assets beta at once

$$\begin{aligned} \beta &= \frac{1}{\sigma_m^2} \cdot \Sigma \cdot w_m \\ &= \frac{1}{0,1205459} \cdot \begin{pmatrix} 0,18 & 0,02 & 0,08 \\ 0,02 & 0,1 & 0,1 \\ 0,08 & 0,1 & 0,25 \end{pmatrix} \cdot \begin{pmatrix} 0,3246753 \\ 0,2294372 \\ 0,4458874 \end{pmatrix} \approx \begin{pmatrix} 0,818786 \\ 0,6140895 \\ 1,3305272 \end{pmatrix} \\ \beta_1 &= 0,818786 \\ \beta_2 &= 0,6140895 \\ \beta_3 &= 1,3305272 \end{aligned}$$

The SML:

$$\bar{r}_i = r_f + \beta_i (\bar{r}_m - r_f)$$

For asset 1

$$r_f + \beta_1 (\bar{r}_m - r_f) = 0,02 + 0,818786(0,1177056 - 0,02) = 0,0999999774016 \approx 0,1 = \bar{r}_1$$

Im thinking the difference is simply inaccuracy of the calculations

Thus the asset is on the SML

f)

we have that in capm

$$\sigma_i^2 = \beta_i \cdot \sigma_m^2 + \sigma_{\epsilon_i}^2 \Leftrightarrow \sigma_i^2 - \beta_i \cdot \sigma_m^2 = \sigma_{\epsilon_i}^2$$

Where $\sigma_{\epsilon_i}^2$ is the unsystematic risk

We simply calculate these

Asset 1

$$0,18 - 0,818786^2 \cdot 0,1205459 \approx 0,0991848$$

Asset 2

$$0,1 - 0,6140895^2 \cdot 0,1205459 \approx 0,0545414$$

Asset 3

$$0,25 - 1,3305272^2 \cdot 0,1205459 \approx 0,0365973$$

Thus asset 1 has the highest unsystematic risk

Exercise 2

a)

We write up a bond market matrix

$$B = \begin{pmatrix} 103 & 0 & 0 \\ 4 & 104 & 0 \\ 6 & 6 & 106 \end{pmatrix}$$

When the calculate discountfactors using the spotrates

$$d_1 = \frac{1}{1 + 0.04} \approx 0,9615385$$

$$d_2 = \frac{1}{(1 + 0.05)^2} \approx 0,9070295$$

$$d_3 = \frac{1}{(1 + 0.07)^3} \approx 0,8162979$$

As a vector

$$d = \begin{pmatrix} 0,9615385 \\ 0,9070295 \\ 0,8162979 \end{pmatrix}$$

Then we can simply get a pricevector

$$B \cdot d = P$$
$$\begin{pmatrix} 103 & 0 & 0 \\ 4 & 104 & 0 \\ 6 & 6 & 106 \end{pmatrix} \begin{pmatrix} 0,9615385 \\ 0,9070295 \\ 0,8162979 \end{pmatrix} \approx \begin{pmatrix} 99,0384655 \\ 98,177222 \\ 97,7389854 \end{pmatrix}$$

With the asset being in the order

A,B,C

b) yes since B is invertible especially there are 3 linearly independent assest being traded and thus any cashflow wanted can be created as a linear combination of the 3 coupon bonds

short: since $Rank(B) = 3 = T$ then the market is complete

c)

we calculate the arbitragefree price of the bond in this market

$$P = B' \cdot d$$
$$B' = (0 \quad 100 \quad 0)$$
$$P = (0 \quad 100 \quad 0) \cdot \begin{pmatrix} 0,9615385 \\ 0,9070295 \\ 0,8162979 \end{pmatrix} = 90,70295$$

Thus the price of the ZCB is to low and there is arbitrage in the market

We will show this further in the next part

d) we replicate the cashflow in the market then “buy low sell high” selling the expensive asset and buying the cheap one

we recreate the cashflow in the market of coupon bonds

$$(0 \quad 100 \quad 0) \begin{pmatrix} 103 & 0 & 0 \\ 4 & 104 & 0 \\ 6 & 6 & 106 \end{pmatrix}^{-1} \approx (-0,0373413 \quad 0,9615385 \quad 0) = \theta$$

This portfolio pays out exactly 100 at time 2 just like the ZCB

And it costs

$$(-0,0373413 \quad 0,9615385 \quad 0) \begin{pmatrix} 99,0384655 \\ 98,177222 \\ 97,7389854 \end{pmatrix} \approx 90,7029537$$

To make a profit we simply sell the replication and buy the ZCB making the total portfolio

With the weights being in the order A,B,C,ZCB

$$\theta_{total} = -(-0,0373413 \quad 0,9615385 \quad 0 \quad 0) + (0 \quad 0 \quad 0 \quad 1) \\ = (0,0373413 \quad -0,9615385 \quad 0 \quad 1)$$

We update our market matrix to make sure we have 0 cashflows

$$B = \begin{pmatrix} 103 & 0 & 0 \\ 4 & 104 & 0 \\ 6 & 6 & 106 \\ 0 & 100 & 0 \end{pmatrix} \\ B_{\theta} = (0,0373413 \quad -0,9615385 \quad 0 \quad 1) \begin{pmatrix} 103 & 0 & 0 \\ 4 & 104 & 0 \\ 6 & 6 & 106 \\ 0 & 100 & 0 \end{pmatrix} = (-1 \cdot 10^{-7} \quad -4 \cdot 10^{-6} \quad 0)$$

Again I believe the small difference is due to computation error

For the price

$$(0,0373413 \quad -0,9615385 \quad 0 \quad 1) \begin{pmatrix} 99,0384655 \\ 98,177222 \\ 97,7389854 \\ 90 \end{pmatrix} \approx -0,7029537$$

Since $B_1, B_2, B_3 \geq 0, P < 0$ there is arbitrage

To take advantage of this we simply go long a huge amount of the portfolio described making 0.7 for every one

e)

plugging into the formula

$$s_t = 0.05 - 0.01 \cdot \frac{1 - e^{-t}}{t} + 0.04 \cdot \left(\frac{1 - e^{-t}}{t} - e^{-t} \right) \\ s_{\frac{1}{2}} = 0.05 - 0.01 \cdot \frac{1 - e^{-\frac{1}{2}}}{\frac{1}{2}} + 0.04 \cdot \left(\frac{1 - e^{-\frac{1}{2}}}{\frac{1}{2}} - e^{-\frac{1}{2}} \right) \approx 0,0493469 \\ s_{\frac{3}{2}} = 0.05 - 0.01 \cdot \frac{1 - e^{-1.5}}{1.5} + 0.04 \cdot \left(\frac{1 - e^{-1.5}}{1.5} - e^{-1.5} \right) \approx 0,0566122 \\ s_{\frac{5}{2}} = 0.05 - 0.01 \cdot \frac{1 - e^{-2.5}}{2.5} + 0.04 \cdot \left(\frac{1 - e^{-2.5}}{2.5} - e^{-2.5} \right) \approx 0,0577316$$

f)

flat price

$$P^{full} = PV(\text{future cash flows}) = P^{flat} + AI.$$

$$AI = \frac{t - T_1}{T_2 - T_1} \frac{cF}{m}.$$

Since we are exactly one year from the next payment

Then we are at $t = T_1$

Thus

Since we are exactly one year before delivery there is no accrued interest and the flat price is simply the price

$$AI = 0$$

And

$$P_{flat} = PV(\text{future cashflows}) = P_C = 97,738,985.4$$

exercise 3

we note a couple of things in the beginning

first the model has constant up/downs

$$u = 1.1$$

$$d = 0.9$$

A continuous compounded risk free rate and thus a discount factor

$$r = 0.02$$

$$d_{0,t} = e^{-r \cdot t}$$

And that the derivative described is a combination of a european put and call that can have different strikes that I will note

$$C^{X_1} - P^{X_2}$$

a)

we simply have a formula

$$q = \frac{e^{r \cdot \Delta t} - d}{u - d} = \frac{e^{0.02} - 0.9}{1.1 - 0.9} \approx 0,6010067$$
$$1 - q = 1 - 0,6010067 = 0,3989933$$

We know that a property of the risk neutral measure and pricing in general is

$$S_0 = E^Q[S_T] \cdot d_{0,T} \Rightarrow E^Q[S_T] = \frac{S_0}{d_{0,T}}$$

I will still do the calculation of course

Im thinking we are to calculate

$$E^Q[S_1], E^Q[S_2], E^Q[S_3]$$

$$E^Q[S_1] = q \cdot S_u + (1 - q) \cdot S_d = 0,6010067 \cdot 110 + 0,3989933 \cdot 90 = 102,020134$$

$$E^Q[r_1] = \frac{E^Q[S_1]}{S_0} - 1 = \frac{102,020134}{100} - 1 \approx 0,0202013 = e^{0.02} - 1$$

$$E^Q[S_2] = q^2 \cdot S_{uu} + 2 \cdot (q \cdot (1 - q)) \cdot S_{ud} + (1 - q)^2 \cdot S_{dd}$$
$$= 0,6010067^2 \cdot 121 + 2 \cdot 0,6010067 \cdot 0,3989933 \cdot 99 + 0,3989933^2 \cdot 81$$
$$\approx 104,0810774$$

$$E^Q[r_2] = \frac{104,0810774}{100} - 1 \approx 0,0408108 = e^{0.04} - 1$$

$$E^Q[S_3] = q^3 \cdot S_{uuu} + 3 \cdot q^2 \cdot (1 - q) \cdot S_{uud} + 3 \cdot q \cdot (1 - q)^2 \cdot S_{udd} + (1 - q)^3 \cdot S_{ddd}$$
$$= 0,6010067^3 \cdot 133.1 + 3 \cdot 0,6010067^2 \cdot 0,3989933 \cdot 108.9 + 3 \cdot 0,6010067 \cdot 0,3989933^2 \cdot 89.1$$
$$+ 0,3989933^3 \cdot 72.9 \approx 106,1836546$$

$$E^Q[r_3] = \frac{106,1836546}{100} - 1 \approx 0,0618365 = e^{0.06} - 1$$

b)

this is simply a forward contract

by realising our position is a combination of a european put and a call in this situation with the same strike we can use the put call parity since we are long the call and short the put with same strike and expiration which says that the price of this must be the price of a forward with strike K

$$C - P = S_0 - K \cdot e^{-rT}$$

$$= 100 - 95 \cdot e^{-0.02 \cdot 3} \approx 10,5323693$$

The payout can be written

$$\max\{0, S_3 - K\} - \max\{K - S_3, 0\} = S_3 - K$$

Which is the payoff of a forward contract

c) the derivative is still just a combination of a call and a put thus we price them separately and calculate add them together

first the call

$$C^{105}$$

We note that it only pays out in the “uuu” state and “uud” states

And price using the expected value under Q and discounting it

$$C_0^{105} = e^{-0.02 \cdot 3} \cdot (q^3 \cdot (133.1 - 105) + 3 \cdot q^2 \cdot (1 - q) \cdot (108.9 - 105))$$

$$= e^{-0.02 \cdot 3} \cdot (0,6010067^3 \cdot (133.1 - 105) + 3 \cdot 0,6010067^2 \cdot (1 - 0,6010067) \cdot (108.9 - 105))$$

$$\approx 7,3329615$$

Then the put

Which again only pays out in the “ddd” and “udd” states and do the exact same as with the call

$$P_0^{95} = e^{-0.02 \cdot 3} \cdot (q \cdot (1 - q)^2 \cdot (95 - 89.1) \cdot 3 + (1 - q)^3 \cdot (95 - 72.9))$$

$$= e^{-0.02 \cdot 3} \cdot (0,6010067 \cdot (1 - 0,6010067)^2 \cdot (95 - 89.1) \cdot 3 + (1 - 0,6010067)^3 \cdot (95 - 72.9))$$

$$\approx 2,9168729$$

Thus the value of the derivative

$$P_0^d = C_0^{105} - P_0^{95} = 7,3329615 - 2,9168729 = 4,4160886$$

d) again we simply calculate the expected value under Q and discount

we note that it pays out 10 in the “uuu” and “uud” states

$$P_0^{binary} = e^{-0.02 \cdot 3} \cdot (10 \cdot (q^3 + 3(q^2 \cdot (1 - q))))$$

$$= e^{-0.02 \cdot 3} \cdot (10 \cdot (0,6010067^3 + 3(0,6010067^2 \cdot (1 - 0,6010067)))) \approx 6,1162807$$

The binary is more expensive than the asset in (c) but has no downside risk it misses out on the bigger upside in the “uuu” case but earns more in the “uud” state which is more likely

Trying to relate the asset to the asset from (b) we note that it is cheaper and again has no downside risk but the forward has bigger upsides in both states

Im thinking this asset might be good as an alternative for a call with lower variance of return but also lower upside potential (no probability of hitting it out of the park)

e) at all times we simply solve the system of equations with “0” being the state we are in and “u”, “d” being going up and down from there

$$P_u = S_u \cdot \Delta + e^{r \cdot \Delta t} \cdot B$$

$$P_d = S_d \cdot \Delta + e^{r \cdot \Delta t} \cdot B$$

Where Δ is the amount of the asset in we need to hold

And B is the amount we need to invest in the risk free asset

“uu”

$$10 = 133.1 \cdot \Delta + e^{0.02} \cdot B$$

$$10 = 108.9 \cdot \Delta + e^{0.02} \cdot B$$

$\Updownarrow$

$$\Delta = 0 \quad \wedge \quad B = 9,8019867$$

Makes sence since we have a guarentied return of 10

“ud”

$$10 = 108.9 \cdot \Delta + e^{0.02} \cdot B$$

$$0 = 89.1 \cdot \Delta + e^{0.02} \cdot B$$

$\Updownarrow$

$$\Delta = 0,5050505 \quad \wedge \quad B = -44,1089403$$

“dd”

$$0 = 89.1 \cdot \Delta + e^{0.02} \cdot B$$

$$0 = 72.9 \cdot \Delta + e^{0.02} \cdot B$$

$\Updownarrow$

$$\Delta = 0 \quad \wedge \quad B = 0$$

Which again makes sence since we never get a payout

f)

again we simply find the expected value under Q and discount it

we note that it pays out if $S_T \geq K$

thus we can write the payoff

$$= 10 \cdot \mathbb{I}_{\{S_T \geq K\}}$$

Since we are in a BSM world we know from slides that

$$d \log(S_T) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$$

And that under the risk neutral measure using girsanows theorem

$$d \log(S_T) = \left(r - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t^Q$$

This also means

$$\begin{aligned} \log(S_T) &= \log(S_t) + \int_t^T \left(r - \frac{1}{2} \sigma^2 \right) ds + \int_t^T \sigma dW_s^Q \\ &= \log(S_t) + (T - t) \cdot \left(r - \frac{1}{2} \sigma^2 \right) + \sigma \cdot W_{T-t} \end{aligned}$$

Where

$$W_{T-t} \sim N(0, T-t)$$

Thus $S_T = S_t \cdot e^{\left(r - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma \cdot W_{T-t}}$

We now calculate

The expectation under Q

$$E^Q[10 \cdot \mathbb{I}_{\{S_T \geq K\}}]$$

We note that

$$E^Q[10 \cdot \mathbb{I}_{\{S_T \geq K\}}] = 10 \cdot P^Q(S_T \geq K)$$

$$\begin{aligned} S_T \geq K &\Leftrightarrow S_t \cdot e^{\left(r - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma \cdot W_{T-t}} \geq K \\ \left(r - \frac{\sigma^2}{2}\right) \cdot (T-t) + \sigma \cdot W_{T-t} &\geq \log\left(\frac{K}{S_t}\right) \\ \sigma \cdot W_{T-t} &\geq \log\left(\frac{K}{S_t}\right) - \left(r - \frac{\sigma^2}{2}\right) \cdot (T-t) \end{aligned}$$

We introduce a variable

$$X = \frac{W_{T-t}}{\sqrt{T-t}} \sim N(0,1)$$

$$\begin{aligned} \sigma \cdot W_{T-t} &= \sigma \cdot \sqrt{T-t} \cdot X \geq \log\left(\frac{K}{S_t}\right) - \left(r - \frac{\sigma^2}{2}\right) \cdot (T-t) \\ \Leftrightarrow X &\geq \frac{\log\left(\frac{K}{S_t}\right) - \left(r - \frac{\sigma^2}{2}\right) \cdot (T-t)}{\sigma \cdot \sqrt{T-t}} = -d_2 \end{aligned}$$

We note that the expression above is simply $-d_2$ as given in the exercise

$$P^Q(S_T \geq K) = P^Q(X \geq -d_2) = 1 - P^Q(X \leq -d_2)$$

Since X is standard normal

$$= 1 - N(-d_2)$$

But again since its standard normal

$$= N(d_2)$$

Thus

$$E^Q[10 \cdot \mathbb{I}_{\{S_T \geq K\}}] = 10 \cdot N(d_2)$$

And we simply need to discount it

$$p^{binary} = e^{-r(T-t)} \cdot E^Q[10 \cdot \mathbb{I}_{\{S_T \geq K\}}] = e^{-r(T-t)} \cdot 10 \cdot N(d_2)$$

Just as given in the exercise thus her claim must be right

Exercise 4

a)

We use general pricing principles

$$P = E^Q[d_1 \cdot X]$$

In the first period the interestrate is certain and that it only pays out 100 in the “down state”

$$P = \frac{1}{1.05} \cdot (1 - q) \cdot 100 = \frac{1}{1.05} \cdot (1 - 0.5) \cdot 100 \approx 47,6190476$$

b)

we calculate the stateprices on a tree in excel and add them up to get the ZCB prices using the formulas

The binomial model: The forward equation (cont'd)

Theorem (Forward equation in binomial model)

In a binomial model, the state price $\psi_{t,s}$ can be calculated according to

$$\psi_{t,s} = \begin{cases} \frac{(1-q)}{1+s_{t-1,t}(s)} \psi_{t-1,s}, & \text{if } s = 0 \\ \frac{(1-q)}{1+s_{t-1,t}(s)} \psi_{t-1,s} + \frac{q}{1+s_{t-1,t}(s-1)} \psi_{t-1,s-1}, & \text{if } 0 < s < t \\ \frac{q}{1+s_{t-1,t}(s-1)} \psi_{t-1,s-1}, & \text{if } s = t \end{cases}$$

Proof on board

Zero-coupon bond prices can then be found as

$$B_{0,T} = \sum_{s=0}^T \psi_{T,s}$$

22 / 45

This gives us the the stateprice tree

state prices						
	t=0		t=t1		t=t2	t=t3
			1,000000		2,000000	3,000000
				"uu"	0,224618	0,207980
		"u"	0,476190	"ud"	0,224618	0,213922
	1,000000				0,455779	0,434075
		"d"	0,476190	"du"	0,231160	0,220153
				"dd"	0,231160	0,228872
ZCB Prices						
	B_0,0		B_0,1		B_0,2	B=0,3
	1,000000		0,952381		0,911557	0,870926

With tme middel row after the first denoting where the 2 paths meet adding them together

The sum of these are exactly our ZCB prices with principal 1

$$B_{0,1} = 100 \cdot 0.952381 = 95,2381$$

$$B_{0,2} = 100 \cdot 0.911557 = 91,1557$$

$$B_{0,3} = 0.870926 \cdot 100 = 87,0926$$

The yields on ZCB are exactly the spotrates

$$\frac{100}{1 + \lambda_{0,1}} = 95,2381$$



$$\lambda_{0,1} = 0,0499999 = 5\%$$

$$\lambda_{0,2}: \frac{100}{(1 + \lambda)^2} = 91,1557$$



$$\lambda = 0,0473892$$

This one is slightly off im hoping that there is a missing didget since the 2 others are spot on 😊

$$\lambda_{0,3}: \frac{100}{(1 + \lambda)^3} = 87,0926$$



Ligningen løses for λ vha. WordMat.

$$\lambda = 0,0471436$$

c) the price of the callable bond

$$P_{\text{callable}} = P_{\text{noncallable}} - P_{\text{option}}$$

We will simply price the option

this person will call the bond if the the value of the bond is higher than 90

thus if he can make a profit at time 1 he will buy it back for 90 and im assuming sell it again simply pocketing the difference

we calculate the ZCB prices in the “up state” this we again have done on a tree where we calculate the ZCB prices backwards with the formula

Definition (Zero-coupon bond prices)

In the binomial model of stochastic interest rates, we denote by $B_{t,T}^s$ the zero coupon bond price at time t in state s of a zero coupon bond with maturity T .

A zero coupon bond with maturity $T_i \in (0, T]$ always has

$$B_{T_i,T_i}^s = 1$$

Working backwards we then have:

$$B_{t-1,T_i}^s = \frac{1}{1 + s_{t-1,t}} \mathbb{E}^{\mathbb{Q}} [B_{t,T_i}^s] = \frac{1}{1 + s_{t-1,t}} \left[q B_{t,T_i}^{s+1} + (1 - q) B_{t,T_i}^s \right]$$

Repeat all the way back to $t = 1, \dots, T_i$ and $s = 0, \dots, t - 1$.

ZCB prices	3 period ZCB				
	t=0	t=t1	t=t2	t=t3	
Bond prices in all states of the world	0,000000	1,000000	2,000000	3,000000	
			0,925926		
		0,885994			
	0,870926		0,952381	1,000000	
		0,942951			
			0,990099		

The value of the ZCB with expiration at time 3 can then easily be calculated

$$B_{1,3}^u = 0.885884 \cdot 100 \approx 88,5884$$

$$B_{1,3}^d = 100 \cdot 0.942951 = 94,2951$$

We then realise that he exercises in the “down” state

Earning

$$94.2951 - 90 \approx 4,2951$$

We can price this option with state prices

$$\psi_{1,0} = \frac{0.5}{1.05} \approx 0,4761905$$

$$4.2951 \cdot 0,4761905 \approx 2,0452858$$

Then we price the callable bond

$$P_{callable} = 87,0926 - 2,0452858 = 85,0473142$$

d)

we need to have enough of the claim that we get 4,2951 in state “d”

thus

$$\frac{4,2951}{100} \approx 0,042951$$

This costs

$$47,6190476 \cdot 0,042951 \approx 2,0452857$$

As well as our normal bond

$$B_{03} = 87,0926$$

By selling the bond and buying the derivative we reach the position of the callable bond

The price we reach is then

$$P_{portfolio} = 87,0926 - 2,0452857 \approx 85,0473143$$

Which is (with a very small error) the same as the price of the callable bond

e)

e. A friend of yours has suggested that instead of modelling short-term interest rates via a binomial tree, you should consider a “2-factor Vasicek” model of the form

$$s_t = X_t + aY_t,$$

where a is a constant, and the dynamics of X_t and Y_t are given by the Ornstein-Uhlenbeck processes

$$dX_t = \kappa_X (\theta_X - X_t) dt + \sigma_X dW_t^{\mathbb{Q}}$$

and

$$dY_t = \kappa_Y (\theta_Y - Y_t) dt + \sigma_Y dZ_t^{\mathbb{Q}},$$

with $W_t^{\mathbb{Q}}$ and $Z_t^{\mathbb{Q}}$ independent Brownian motions under $\mathbb{Q}$. Show that s_t also follows an Ornstein-Uhlenbeck process if and only if $\kappa_X = \kappa_Y$. Express the long-term mean and volatility of s_t under this assumption.

We write both out

$$\begin{aligned} X_T &= X_t + \int_t^T \kappa_X (\theta_X - X_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} \Rightarrow X_T - X_t = \int_t^T \kappa_X (\theta_X - X_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} \\ Y_T &= Y_t + \int_t^T \kappa_Y (\theta_Y - Y_t) dt + \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \Rightarrow Y_T - Y_t = \int_t^T \kappa_Y (\theta_Y - Y_t) dt + \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \\ s_T - s_t &= X_T - X_t + \alpha \cdot (Y_T - Y_t) \\ &= \int_t^T \kappa_X (\theta_X - X_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} + \alpha \cdot \left(\int_t^T \kappa_Y (\theta_Y - Y_t) dt + \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \right) \\ &= \int_t^T \kappa_X (\theta_X - X_t) + \alpha \kappa_Y (\theta_Y - Y_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} + \alpha \cdot \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \end{aligned}$$

If and only if $\kappa_X = \kappa_Y$ then we call it κ_s

$$\begin{aligned} &= \int_t^T \kappa_X (\theta_X - X_t) + \alpha \kappa_Y (\theta_Y - Y_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} + \alpha \cdot \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \\ &= \int_t^T \kappa_s (\theta_X + \alpha \theta_Y - X_t - \alpha Y_t) dt + \sigma_X \cdot W_{T-t}^{\mathbb{Q}} + \alpha \cdot \sigma_Y \cdot Z_{T-t}^{\mathbb{Q}} \end{aligned}$$

Since W and Z are independent normal the sum is normal

$$\begin{aligned} E[\sigma_X W + \sigma_Y \alpha Z] &= 0 \\ V[\sigma_X W + \sigma_Y \alpha Z] &= \sigma_X^2 \cdot (T - t) + \alpha^2 \cdot \sigma_Y^2 \cdot (T - t) \end{aligned}$$

Thus

$$\sigma_X W + \sigma_Y \alpha Z \sim N(0, (\sigma_X^2 + \alpha^2 \cdot \sigma_Y^2)(T - t))$$

By introducing a new normal variable

$$G_{T-t} \sim N(0, (T - t))$$

And setting

$$\sigma_S = \sqrt{\sigma_X^2 + \alpha^2 \cdot \sigma_Y^2}$$

We can write

$$= \int_t^T \kappa_s (\theta_X + \alpha \theta_Y - X_t - \alpha Y_t) dt + \sigma_S G_{T-t}$$

My setting

$$\theta_S = \theta_X + \alpha \cdot \theta_Y$$

We get

$$= \int_t^T \kappa_s (\theta_S - X_t - \alpha Y_t) dt + \sigma_S G_{T-t} = \int_t^T \kappa_s (\theta_S - s_t) dt + \sigma_S G_{T-t}$$

Thus it is a Ornstein-Uhlenbeck process

If $\kappa_X \neq \kappa_Y$ we can't do the proper manipulations inside the integral and thus it can not become an Uhlenbeck process

$$\mathbb{E}[r_T | r_t] = \theta + (r_t - \theta)e^{-\kappa(T-t)}, \quad \mathbb{V}^{\mathbb{Q}}[r_T | r_t] = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa(T-t)})$$

$$E[s_T \text{ when } s_t] = \theta_S + (s_t - \theta_S) \cdot e^{-\kappa_S(T-t)}$$

$$V[s_T \text{ when } s_t] = \frac{\sigma_S^2}{2\kappa_S} (1 - e^{-2 \cdot \kappa_S(T-t)})$$

f. Using Ito's lemma, express the dynamics of $f(t, X_t) = e^{\kappa X_t} X_t$, and explain why this transformation can be useful if we want to calculate $\mathbb{E}^{\mathbb{Q}}[X_T | X_t]$, $\mathbb{E}^{\mathbb{Q}}[Y_T | Y_t]$, and $\mathbb{E}^{\mathbb{Q}}[s_T | s_t]$ for any $T \geq t$ in the model described in part e).

Ito's lemma

$$df(t, X(t)) = \left\{ \frac{\partial f}{\partial t}(t, X(t)) + a(t, X(t)) \frac{\partial f}{\partial x}(t, X(t)) + \frac{1}{2} b(t, X(t))^2 \cdot \frac{\partial^2 f}{\partial x^2}(t, X(t)) \right\} dt$$

$$+ b(t, X(t)) \cdot \frac{\partial f}{\partial x}(t, X(t)) dW_t$$

$$\frac{\partial f}{\partial t} = \kappa \cdot f$$

$$\frac{\partial f}{\partial X} = \frac{f}{X}$$

$$\frac{\partial^2 f}{\partial X^2} = 0$$

$$a = \kappa \cdot (\theta - X)$$

$$\begin{aligned}
 b &= \sigma \\
 df &= \left(\kappa \cdot f + \kappa \cdot (\theta - X) \cdot \frac{f}{X} \right) dt + \sigma \cdot \frac{f}{X} \cdot dW_t \\
 &= (\kappa \cdot e^{\kappa \cdot t} \cdot X + \kappa \cdot \theta \cdot e^{\kappa \cdot t} - \kappa \cdot e^{\kappa \cdot t} \cdot X) dt + \sigma \cdot e^{\kappa \cdot t} dW_t \\
 &= (\kappa \cdot \theta \cdot e^{\kappa \cdot t}) dt + \sigma \cdot e^{\kappa \cdot t} dW_t
 \end{aligned}$$

This is quite nice because we no longer have to solve any stochastic integrals thus to get an expression with W_{T-t} which is just normal then it becomes way easier to calculate the expectation by using Ito's lemma

Under is just an expression of the function f

$$f(s_T, T) = f(S_t, t) + \int_t^T (\kappa \cdot \theta \cdot e^{\kappa \cdot s}) ds + W_{T-t}$$

